Sri Lanka Institute of Information Technology



Lab Submission - 03

< 2026-Y2-S1-KU-36 >

No.	Member	Registration No.	Assigned Function
1	S M H C B D Rambukpitiya	IT25103736	User & Customer Management
2	I K S P Jayawardana	IT25103732	Vehicle Management
3	W M K K G N A Weerasingha	IT25103737	Booking & Reservation Management
4	R M M D Bhagyamathi	IT25103733	Payment Management
5	R M R D Rajapaksha	IT25103734	Feedback & Support Management
6	K A K Dananjaya	IT25103735	Fleet Management

<RideLanka FleetHire>

< Web-based Car Rental Management System >

Software Engineering | SE2030

B.Sc. (Hons) in Information Technology


SE2030 - SOFTWARE ENGINEERING
YEAR 02 - SEMESTER 01
PROPOSAL REPORT
RIDELANKA FLEETHIRE SYSTEM
Web based Car Rental Management System
RIDE LANKA
FLEETHIRE
MODULE
SE2030
SOFTWARE ENGINEERING
GROUP
2026-Y2-S1-KU-36
SIX MEMBERS - SIX CORE FUNCTIONS
SUBMISSION DATE
18 AUGUST 2026
GROUP MEMBERS AND CORE OWNERSHIP
01 S M H C B D Rambukpitiya - IT25103736
User & Customer Management
02 I K S P Jayawardana - IT25103732
Vehicle Management
03 W M K K G N A Weerasingha - IT25103737
Booking & Reservation Management
04 R M M D Bhagyamathi - IT25103733
Payment Management
05 R M R D Rajapaksha - IT25103734
Feedback & Support Management
06 K A K Dananjaya - IT25103735
Fleet Management

01 Report Overview

RideLanka FleetHire is a web-based car rental management system connecting vehicle search, online booking, verified payment, digital receipts, fleet and maintenance operations, customer feedback handling, and role-controlled access through one shared platform.

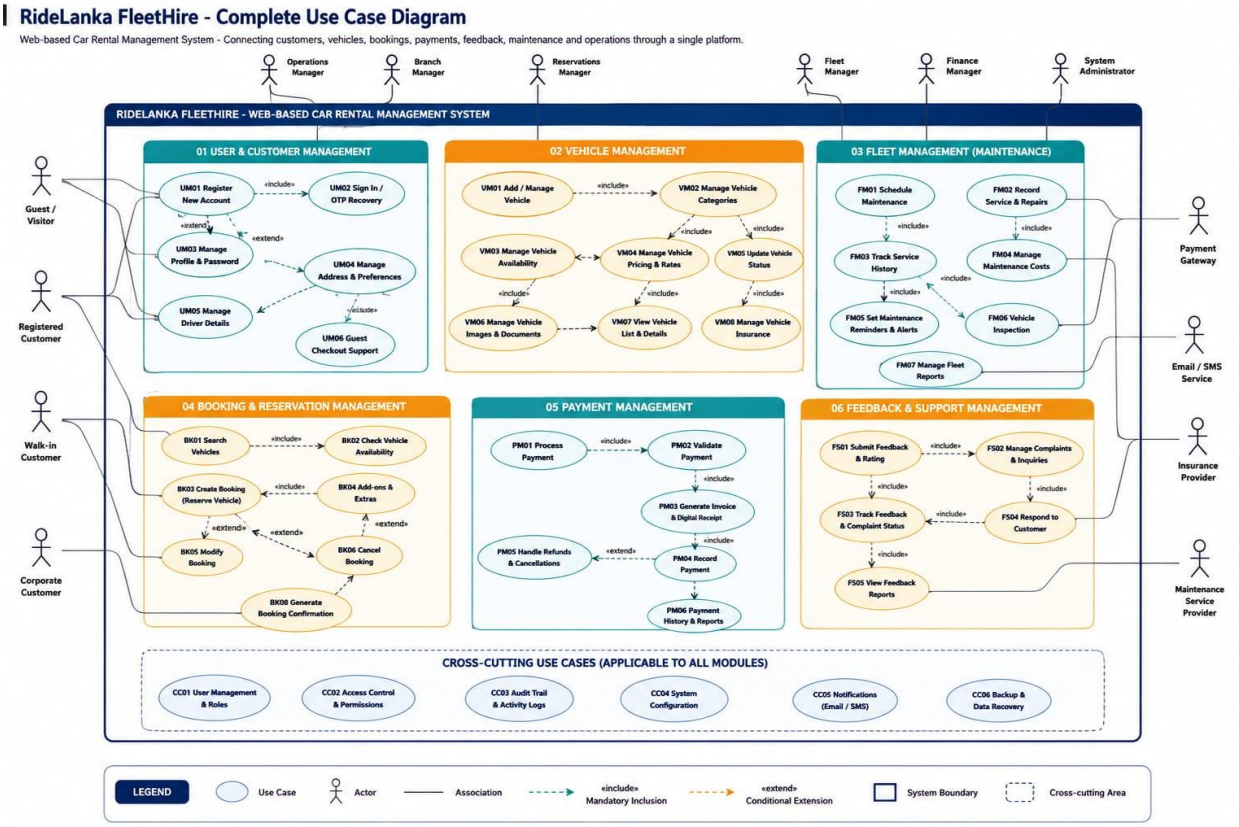
Group Details and Core Ownership

No.	Member	Registration No.	Assigned Function
1	S M H C B D Rambukpitiya	IT25103736	User & Customer Management
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OWNERSHIP CONTINUITY These six function owners are the original CRUD responsibility division confirmed in the project proposal (Section 6, Six Major Functions). Each member's scenario below traces to the same function.

02 RideLanka FleetHire - Complete Use Case Diagram

The diagram below groups the six member-owned CRUD functions inside the RideLanka FleetHire system boundary, with human actors and external systems (Payment Gateway, Email/SMS Service, Insurance Provider) shown at the edges. Dashed teal arrows denote <<include>> relationships; dashed orange arrows denote <<extend>> relationships.



01 User & Customer Management Use Case Scenario

UM01 - Register a New Customer Account

OWNERSHIP AND TRACEABILITY S M H C B D Rambukpitiya - IT25103736 | Core ownership: User & Customer Management | Diagram use case: UM01

Use Case Definition

Use case ID	UM01
Use case name	Register a New Customer Account
Goal in context	Let a new customer create a verified, least-privilege account so they can search, book, pay for, and manage vehicle rentals online.
Scope	RideLanka FleetHire - User & Customer Management
Level	User goal
Summary	A visitor supplies identity and contact details, sets a password, and verifies the account through an OTP before it becomes usable for booking.
Priority	Critical - gateway to every customer-facing transaction
Primary actor	Customer (unregistered visitor)
Supporting actors	Email / SMS Service
Stakeholders and interests	Customer needs a fast, low-friction sign-up; Reservations and Finance need a verified identity linked to every booking and payment; Administration needs no duplicate or fraudulent accounts.
Trigger	The visitor selects Create Account from the RideLanka FleetHire web application.
Preconditions	The visitor is not already signed in; a unique email or mobile number, name, and a compliant password can be supplied; the OTP channel is reachable.
Success postcondition	One ACTIVE customer account exists with a verified contact channel, a securely stored password, and the default customer role, ready to search and book vehicles.
Failure guarantee	No duplicate, unverified, or partially created account remains; no password is stored in plain text.
Related use cases	UM02 Sign In / OTP Recovery; UM03 Manage Profile & Password; UM06 Guest Checkout Support; BR02 Create Online Booking.
Implementation alignment	Aligned to the current self-registration flow with OTP verification before first booking. A social-login option is not yet implemented.

Main Success Scenario

Step	Actor / System Interaction
1	The visitor opens Create Account and enters full name, email or mobile number, and a password.
2	The system validates the name, contact format, and password strength.
3	The system checks that the email or mobile number is not already registered.
4	The system protects the password before storing the new account as PENDING_VERIFICATION.
5	The system sends a one-time verification code to the supplied email or mobile number.
6	The visitor enters the received one-time code.
7	The system verifies the code against the stored value and expiry window.
8	The system activates the account with the default Customer role.
9	The system records account creation and verification in the audit history.
10	The system signs the customer in and displays the account home page.

Alternative and Exception Flows

Ref	Condition	System Response
2a	Contact detail or password fails validation	The system explains the specific rule and keeps the account unsaved.
3a	Email or mobile number already registered	The system reports the conflict and offers the sign-in or password-recovery path.
5a	OTP delivery is temporarily unavailable	The account remains PENDING_VERIFICATION and the visitor may request a retry once the channel recovers.
7a	One-time code is incorrect or expired	The system rejects the code, allows a limited number of retries, and offers to resend a new code.
7b	Retry limit is exceeded	The system temporarily locks verification for the account and directs the visitor to request a new code after a cooldown.
9a	Account activation cannot be completed	The registration is cancelled so no unverified or partially active account remains.

Business Rules

- Every account must have a unique verified email or mobile number.
- An account may not book or pay until it is ACTIVE and verified.
- Passwords are never stored or logged in plain text.
- A newly registered account receives exactly the default Customer role.
- Account creation and verification events are retained for audit.

Special Requirements and Open Decisions

Security: rate-limit OTP requests and lock repeated failed attempts. Privacy: contact details are used only for verification and booking communication. Usability: registration should complete in three or fewer screens. Open issues: social-login integration, mobile-number-only markets, and final OTP expiry duration.

02 Vehicle Management Use Case Scenario

VM04 - Mark a Vehicle Out of Service

OWNERSHIP AND TRACEABILITY I K S P Jayawardana - IT25103732 | Core ownership: Vehicle Management | Diagram use case: VM04

Use Case Definition

Use case ID	VM04
Use case name	Mark a Vehicle Out of Service
Goal in context	Prevent an unsafe, damaged, or unavailable vehicle from being searched or booked while preserving an accountable status history.
Scope	RideLanka FleetHire - Vehicle Management
Level	User goal
Summary	Fleet operations staff change an existing vehicle to OUT_OF_SERVICE through a controlled, version-checked status transition that also flags any affected active bookings.
Priority	High - safety, compliance, and booking-inventory accuracy
Primary actor	Fleet Operations Staff
Supporting actors	Maintenance Coordinator
Stakeholders and interests	Fleet staff need an accurate registry; Booking needs unavailable vehicles excluded from search; customers need to be notified if their reservation is affected; auditors need a traceable reason and time.
Trigger	Fleet operations staff determine that a vehicle must be withdrawn from active rental use.
Preconditions	The staff member is signed in and active; the vehicle exists; its current version is displayed; a reason can be supplied; the vehicle is not already OUT_OF_SERVICE.
Success postcondition	The vehicle is OUT_OF_SERVICE, the reason and status history are retained, the vehicle is excluded from future availability search, and any active booking against it is flagged for reassignment.
Failure guarantee	The vehicle remains in its last valid state; no unrecorded or partially applied status change is left behind.
Related use cases	VM01 Register & Categorize Vehicle; VM02 View Fleet & Availability; VM05 Transfer Vehicle Between Branches; FM06 Notify Affected Bookings.
Implementation alignment	Aligned to the current fleet-status command used by front-desk and fleet staff. Automatic reassignment of an affected booking to a replacement vehicle is planned but not yet implemented end to end.

Main Success Scenario

Step	Actor / System Interaction
1	The staff member opens the selected vehicle record.
2	The system displays the current status, category, branch, version, and recent status history.
3	The staff member chooses Mark Out of Service and enters the reason.
4	The system verifies that the actor is active and authorised for fleet status changes.
5	The system confirms that the vehicle still exists and that the displayed version is current.
6	The system confirms that the requested transition to OUT_OF_SERVICE is valid from the current state.
7	The system changes the vehicle status to OUT_OF_SERVICE.
8	The system records the old status, new status, reason, actor, and time in the status history.
9	The system checks for any active or upcoming booking against the vehicle and flags it for staff follow-up.
10	The system shows the updated vehicle and excludes it from future availability search results.

Alternative and Exception Flows

Ref	Condition	System Response
3a	Reason is missing	The system identifies the required field and keeps the vehicle unchanged.
4a	Actor is inactive or unauthorised	The system denies the action without revealing restricted fleet data.
5a	Vehicle no longer exists	The system reports that the vehicle cannot be found and makes no change.
5b	Another user changed the vehicle concurrently	The system reports a version conflict and asks the staff member to reload the latest record.
6a	Vehicle is already OUT_OF_SERVICE	The system rejects the repeated transition and displays the current state.
9a	An active booking is linked to the vehicle	The booking is flagged PAYMENT_REVIEW-equivalent for staff to reassign, upgrade, or refund under FM06.
8a	The status or history change cannot be saved	The complete operation is cancelled and the prior vehicle state remains authoritative.

Business Rules

- Only active fleet operations or maintenance staff may change vehicle status.
- A reason and the current record version are mandatory for the transition.
- An OUT_OF_SERVICE vehicle cannot appear in availability search or new bookings.
- The old state, new state, reason, actor, and time must remain traceable.
- An affected active booking must be flagged, not silently cancelled.

Special Requirements and Open Decisions

Reliability: the status change and history record must either both succeed or both remain unchanged. Usability: the version-conflict message must clearly guide a reload. Open issues: automatic replacement-vehicle suggestion, insurance-provider notification, and a formal incident-linkage workflow.

03 Booking & Reservation Management Use Case Scenario

BR02 - Create an Online Booking and Confirm Reservation

OWNERSHIP AND TRACEABILITY W M K K G N A Weerasingha - IT25103737 | Core ownership: Booking & Reservation Management | Diagram use case: BR02

Use Case Definition

Use case ID	BR02
Use case name	Create an Online Booking and Confirm Reservation
Goal in context	Secure the customer's selected vehicle for the requested dates, apply any valid promotion, complete payment, and issue a confirmed reservation without double booking.
Scope	RideLanka FleetHire - Booking & Reservation Management
Level	User goal
Summary	A registered customer selects an available vehicle and rental period, optionally applies a discount code, supplies pickup and drop-off details, pays through the payment gateway, and receives a confirmed booking.
Priority	Critical - core customer and revenue flow
Primary actor	Registered Customer
Supporting actors	Payment Gateway; Email / SMS Service
Stakeholders and interests	Customer needs one reliable reservation; front-desk staff need accurate vehicle inventory; Finance needs a traceable payment; Reservations needs promotion rules correctly applied.
Trigger	The customer selects an available vehicle from search results and chooses to book it.
Preconditions	The customer is signed in; the vehicle is AVAILABLE for the requested branch and dates; requested add-ons and extras are valid; required customer and driver details can be supplied.
Success postcondition	One confirmed booking owns the selected vehicle for the reserved period, payment is SUCCESS, a digital receipt and confirmation are issued, and a booking-confirmation message is queued.
Failure guarantee	No vehicle is confirmed for two overlapping bookings; no confirmation is issued for an unconfirmed payment; booking and payment outcomes remain traceable.
Related use cases	BR01 Search Vehicle Availability; BR03 Apply Promotion / Discount Code; BR06 Confirm Reservation; PM01 Create Payment / Checkout; VM02 View Fleet & Availability.
Implementation alignment	Core availability check, hold, checkout, gateway callback, and confirmation flow match the executable baseline. Automatic booking-confirmation SMS is a required integration not yet connected end to end.

Main Success Scenario

Step	Actor / System Interaction
1	The customer opens current availability for the selected vehicle, branch, and rental period.
2	The system rechecks the selected vehicle's availability for the requested dates and calculates the base fare.
3	The customer optionally enters a promotion or discount code.
4	The system validates the code and recalculates the fare if it applies.
5	The system creates a time-limited hold on the vehicle for the requested period.
6	The customer supplies driver details, pickup/drop-off branch, and add-ons.
7	The system validates and securely stores the booking details against the hold.
8	The system creates the booking in PENDING_PAYMENT and presents the booking and fare summary.
9	The customer continues to the payment gateway and confirms payment.
10	The payment gateway returns the payment result for the correct order.
11	The system verifies the gateway signature, order, amount, currency, and result.
12	For a successful on-time payment, the system confirms the booking, records payment as SUCCESS, and releases the hold as a confirmed allocation.
13	The system issues a digital receipt and queues the booking-confirmation message.

Alternative and Exception Flows

Ref	Condition	System Response
2a	Vehicle is no longer available for the requested dates	The hold is not created; the system refreshes availability so the customer can choose again.
4a	Promotion code is invalid, expired, or not applicable	The system explains the rule and proceeds with the standard fare.
5a	The vehicle is claimed by another customer first	Only one hold is granted; the losing request is asked to search again.
8a	The hold expires before booking creation	The system rejects the stale hold and requires a new availability check.
11a	Signature, order, amount, or currency is invalid	The system rejects the callback; no booking or receipt is confirmed.
11b	The same callback is received again	The system returns the existing result without creating another booking or charge.
12a	Successful payment arrives after the hold expires	The booking becomes PAYMENT_REVIEW and no confirmation is issued until staff resolve it.
12b	Gateway reports failure or cancellation	Payment becomes FAILED or CANCELLED and the booking remains unconfirmed.
13a	Confirmation message delivery is temporarily unavailable	The confirmed booking remains valid and delivery is retried under a controlled schedule.

Business Rules

- A vehicle may have only one active confirmed booking for a given date range.
- RideLanka FleetHire calculates the fare; a client-supplied total is not authoritative.
- A promotion code must be validated server-side before it can affect the fare.
- A digital receipt is issued only after payment is confirmed as SUCCESS.
- A customer may view and manage only bookings they own.

Special Requirements and Open Decisions

Concurrency: one winner for a same-vehicle, same-date race. Security and privacy: protect driver details, payment evidence, and booking ownership. Accessibility: present availability as both a list and a calendar view. Open issues: final hold duration, guest checkout limits, and cancellation and modification rules.

04 Payment Management Use Case Scenario

PM02 - Verify Payment Result Before Booking Confirmation

OWNERSHIP AND TRACEABILITY R M M D Bhagyamathi - IT25103733 | Core ownership: Payment Management | Diagram use case: PM02

Use Case Definition

Use case ID	PM02
Use case name	Verify Payment Result Before Booking Confirmation
Goal in context	Accept a genuine payment-gateway result exactly once and prevent booking confirmation for an invalid, duplicate, mismatched, failed, or late payment.
Scope	RideLanka FleetHire - Payment Management
Level	Subfunction supporting the booking goal
Summary	The payment gateway submits a callback; RideLanka FleetHire validates it, applies the correct payment and booking outcome, detects duplicate transactions, and retains accepted evidence for reconciliation.
Priority	Critical - financial integrity and booking control
Primary actor	Payment Gateway
Supporting actors	Finance Officer (only when later review or reconciliation is required)
Stakeholders and interests	Customer needs the correct outcome; Finance needs auditable, reconciled evidence; Booking needs a safe confirmation decision; the business must prevent duplicate revenue events.
Trigger	The payment gateway posts a payment callback for an existing RideLanka FleetHire checkout.
Preconditions	The payment and linked booking exist; the payment is normally PENDING; gateway credentials are configured; required callback fields are present.
Success postcondition	A valid on-time success is applied once, payment becomes SUCCESS, the eligible booking is confirmed, a receipt is issued, and accepted callback evidence is retained for reconciliation.
Failure guarantee	An invalid, duplicate, or ineligible result does not confirm a booking or falsely mark a payment successful; its safe outcome is returned or routed for Finance review.
Related use cases	PM01 Create Payment / Checkout; PM03 Detect Duplicate Transaction; PM04 Reconcile Gateway / Bank; PM05 Manage Deposit & Refund; BR02 Create Online Booking.
Implementation alignment	Uses the current payment states and gateway callback fields. It intentionally does not claim an automatically created finance case or a live bank-statement reconciliation feed, since neither exists in the current baseline.

Main Success Scenario

Step	Actor / System Interaction
1	The payment gateway sends merchant ID, order ID, payment ID, amount, currency, status code, and signature.
2	The system validates the required field formats, configured merchant, and callback signature.
3	The system maps the provider status to the corresponding RideLanka FleetHire payment outcome.
4	The system locates the payment by the provider order ID and prevents simultaneous processing of the same payment.
5	The system checks whether the callback represents a duplicate of a result already applied.
6	The system verifies the expected order, amount, currency, and linked booking.
7	For a successful result, the system checks that the linked vehicle hold is still valid.
8	The system confirms the booking, issues a digital receipt, and records payment as SUCCESS.
9	The system retains the accepted callback evidence with the completed outcome for reconciliation.
10	The system returns a safe acknowledgement without exposing protected payment data.

Alternative and Exception Flows

Ref	Condition	System Response
2a	Merchant, format, or signature is invalid	The system rejects the callback before changing any payment or booking state.
3a	Provider reports failed or cancelled	The system records FAILED or CANCELLED and issues no receipt.
3b	Provider reports a chargeback after success	The system records CHARGEBACK for authorised Finance follow-up.
5a	The callback is a duplicate replay	The system returns the existing outcome without repeating any charge or booking action.
6a	Order, amount, or currency does not match	The system rejects the callback and does not invent an automatic finance case.
7a	Successful result arrives after hold expiry	Payment becomes LATE_SUCCESS, the booking is routed to PAYMENT_REVIEW, and no receipt is issued.
8a	The booking or receipt outcome cannot complete	The operation is cancelled so a partial successful result is not reported as final.

Business Rules

- No confirmation may be issued before an accepted successful result is linked to an eligible booking.
- The same provider result must not create duplicate business effects.
- Payment SUCCESS, FAILED, CANCELLED, LATE_SUCCESS, and CHARGEBACK remain distinct outcomes.
- Refund approval and execution are separate authorised Finance actions.
- Accepted provider evidence is retained without later rewriting.

Special Requirements and Open Decisions

Security: protect gateway credentials, verify signatures safely, and redact logs and responses. Reliability: repeated callbacks return one consistent business outcome. Auditability: accepted evidence supports monthly gateway-and-bank reconciliation. Open issues: production credentials, final LATE_SUCCESS disposition, and remaining payment-review acceptance evidence.

05 Feedback & Support Management Use Case Scenario

FS01 - Submit and Resolve a Customer Complaint Ticket

OWNERSHIP AND TRACEABILITY R M R D Rajapaksha - IT25103734 | Core ownership: Feedback & Support Management | Diagram use case: FS01

Use Case Definition

Use case ID	FS01
Use case name	Submit and Resolve a Customer Complaint Ticket
Goal in context	Let a customer raise a complaint or feedback item and have it tracked through assignment, resolution, and closure with a retained satisfaction record.
Scope	RideLanka FleetHire - Feedback & Support Management
Level	User goal
Summary	A customer submits a complaint or feedback tied to a completed or active booking; customer service staff assign, work, and close the resulting ticket, and the outcome feeds satisfaction reporting.
Priority	High - post-rental customer experience and retention
Primary actor	Customer
Supporting actors	Customer Service Supervisor
Stakeholders and interests	Customer needs their issue heard and resolved; customer service needs a clear queue and ownership; management needs satisfaction trends and recurring-issue visibility.
Trigger	The customer opens a booking or rental record and chooses to submit feedback or a complaint.
Preconditions	The customer is signed in; the referenced booking belongs to the customer; a category and description can be supplied.
Success postcondition	A ticket exists with a unique reference, is assigned to a staff member, moves through IN_PROGRESS to RESOLVED and CLOSED, and the outcome is retained for satisfaction reporting.
Failure guarantee	No ticket is silently dropped or closed without a recorded resolution; no customer can view or edit another customer's ticket.
Related use cases	FS02 Assign & Track Ticket; FS03 Resolve & Close Ticket; FS04 Generate Satisfaction Report; FS05 Analyse Recurring Issue Trends.
Implementation alignment	Aligned to the current ticket lifecycle: submission, assignment, resolution, and closure. Automatic escalation rules for overdue tickets are planned but not yet enforced by the system.

Main Success Scenario

Step	Actor / System Interaction
1	The customer selects a booking and chooses Submit Feedback or Raise Complaint.
2	The customer selects a category and enters a description, and optionally a rating.
3	The system validates the required fields and that the booking belongs to the customer.
4	The system creates the ticket in OPEN state with a unique reference number.
5	The system notifies the customer service queue of the new ticket.
6	A customer service staff member reviews the queue and assigns the ticket to themselves or a colleague.
7	The system changes the ticket to IN_PROGRESS and records the assignee.
8	The assigned staff member investigates and records resolution notes and any compensating action.
9	The staff member marks the ticket RESOLVED and the system notifies the customer.
10	The customer confirms the resolution, or the ticket auto-closes after the review window; the system sets the ticket to CLOSED and retains it for reporting.

Alternative and Exception Flows

Ref	Condition	System Response
3a	Booking does not belong to the customer or is invalid	The system rejects the submission and does not create a ticket.
3b	Category or description is missing	The system identifies the required field and keeps the ticket unsaved.
6a	No staff member is available to assign	The ticket remains OPEN and visible in the unassigned queue for supervisor attention.
8a	Investigation requires more information from the customer	The ticket moves to AWAITING_CUSTOMER and resumes on customer reply.
9a	Customer disputes the resolution	The system reopens the ticket to IN_PROGRESS for further handling.
10a	Ticket closure record cannot be saved	The closure is cancelled and the ticket remains in RESOLVED state for retry.

Business Rules

- A customer may raise a ticket only against their own booking.
- A ticket must carry a category, description, and status at every stage.
- Closure requires a recorded resolution outcome.
- A closed ticket may be reopened only through a new dispute action, not direct edit.
- Ticket data feeds satisfaction and recurring-issue reporting without exposing other customers' details.

Special Requirements and Open Decisions

Usability: the customer should be able to track ticket status without contacting staff directly. Privacy: tickets are visible only to their owner and authorised staff. Reliability: status transitions are auditable end to end. Open issues: automatic escalation thresholds, compensation-approval limits, and multi-channel (email/SMS) ticket intake.

06 Fleet Management (Maintenance) Use Case Scenario

FM01 - Schedule and Complete Vehicle Maintenance

OWNERSHIP AND TRACEABILITY K A K Dananjaya - IT25103735 | Core ownership: Fleet Management (Maintenance) | Diagram use case: FM01

Use Case Definition

Use case ID	FM01
Use case name	Schedule and Complete Vehicle Maintenance
Goal in context	Plan, track, and complete a vehicle's maintenance or repair so the vehicle is returned to service safely with a complete, auditable service history.
Scope	RideLanka FleetHire - Fleet Management (Maintenance)
Level	User goal
Summary	A maintenance coordinator schedules a service for a vehicle, the vehicle is automatically marked unavailable for the maintenance window, repair details are logged, and the vehicle status is updated on completion.
Priority	High - vehicle safety and fleet reliability
Primary actor	Maintenance Coordinator
Supporting actors	Fleet Operations Staff; Email / SMS Service
Stakeholders and interests	Maintenance coordinator needs a clear service plan; fleet operations needs vehicles excluded from booking during service; customers with affected bookings need timely notice; auditors need complete service history.
Trigger	The maintenance coordinator identifies that a vehicle is due for scheduled service or requires repair.
Preconditions	The coordinator is signed in and active; the vehicle exists; a service type, date window, and reason can be supplied.
Success postcondition	A maintenance record exists, the vehicle is automatically excluded from booking for the service window, repair details are logged on completion, and the vehicle returns to AVAILABLE with a complete service history entry.
Failure guarantee	No vehicle is left bookable during an open maintenance window; no maintenance record is left incomplete without a final status.
Related use cases	VM04 Mark Vehicle Out of Service; FM02 Log Repair / Service Detail; FM04 Send Maintenance Alert / Notification; FM05 View Complete Service History; FM06 Notify Affected Bookings.
Implementation alignment	Aligned to the current maintenance-scheduling and status-update workflow. Automatic parts-inventory integration and third-party workshop booking are out of scope for the current baseline.

Main Success Scenario

Step	Actor / System Interaction
1	The maintenance coordinator opens the vehicle record and selects Schedule Maintenance.
2	The coordinator enters the service type, scheduled date window, and reason.
3	The system validates the required fields and checks for any conflicting active booking in the window.
4	The system creates the maintenance record in SCHEDULED state.
5	The system automatically marks the vehicle unavailable for the scheduled window.
6	The system sends a maintenance alert to fleet operations and, where a booking conflict exists, queues a customer notification.
7	On the service date, the coordinator marks the maintenance IN_PROGRESS and records work performed, parts used, and cost.
8	The system stores the repair and service details against the maintenance record.
9	On completion, the coordinator marks the maintenance COMPLETED and confirms the vehicle is roadworthy.
10	The system returns the vehicle to AVAILABLE status and appends the entry to the vehicle's complete service history.

Alternative and Exception Flows

Ref	Condition	System Response
3a	Required fields are missing or the window is invalid	The system identifies the issue and keeps the maintenance record unsaved.
3b	An active booking conflicts with the proposed window	The system flags the conflict for FM06 so the booking can be reassigned or the customer notified before scheduling proceeds.
6a	Notification delivery is temporarily unavailable	The maintenance record remains valid and the notification is retried under controlled conditions.
7a	Additional issues are found once work begins	The coordinator extends the service window and reason; affected bookings are re-checked.
9a	The vehicle is found not roadworthy at completion	The system keeps the vehicle OUT_OF_SERVICE and requires a new maintenance record instead of returning it to AVAILABLE.
10a	The completion or history update cannot be saved	The operation is cancelled and the maintenance record remains IN_PROGRESS for retry.

Business Rules

- A vehicle scheduled for maintenance may not be booked for the scheduled window.
- A maintenance record must reach a final status of COMPLETED or CANCELLED; none may be left open indefinitely.
- Repair and service details are retained as part of the vehicle's permanent service history.
- A vehicle returns to AVAILABLE only after an explicit roadworthy confirmation.
- Any active booking affected by scheduled maintenance must be flagged for staff follow-up rather than silently cancelled.

Special Requirements and Open Decisions

Reliability: vehicle-availability changes and maintenance-record changes must present one complete outcome. Usability: alerts should clearly identify the vehicle, window, and reason. Safety: a vehicle failing the roadworthy check cannot return to service without a further completed maintenance record. Open issues: parts-inventory linkage, workshop-partner integration, and predictive maintenance scheduling.